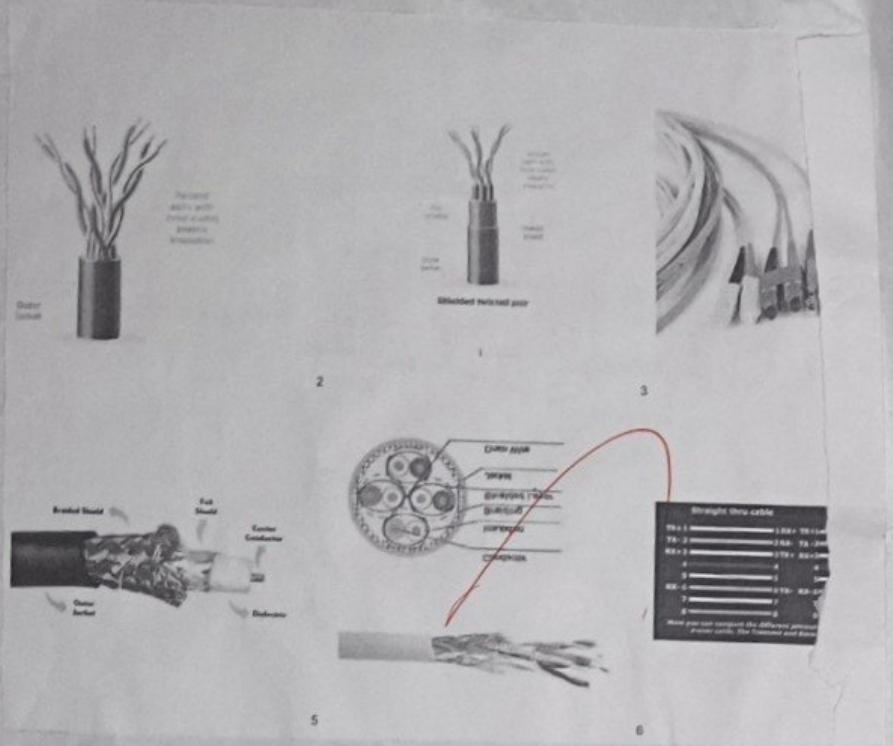
Exp : 2
23/7/26

Study of different types of Networking Cable

AIM:
To Study and understood different types of network Cable and to make our own ethernet (crossover / Straight cable

Theory:

1) Unshielded twisted pair [UTP] Cable.
2) Shielded twisted pair [STP] Cable.
3) Coaxiale Cable
4) Fibre optic Cable.



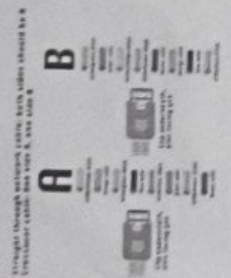
Cable Type	Category	Max data Transmit	Advantages / Disadvantage	Application / use
UTP	3	10bps	Advantage: • cheap in cost • easyle is tall Disadvantage. • more prone to EMI	• 10base - T ethernet • fast ethernet • gigabyte ethernet
	5	upto 100mbps		
	5e	1gbps		
STP	6, 6a	10gbps	Advantage • Sheilded • faster ethernet UTP Disadvantage • Exponisive • Greater instaration effort	• gigabit ethernet 10ab ethernet • widely used in data ce • Gigabest ethern 10 a ethernet
	7	10Gbps		













Coaxial Cable	RG1-6 Ra-5a Ra-11	10-100 mbps	Advantage:) High bandwidth) Immune to interface) Loss is low Disadvantage:) Cost) Size B bulky	Speed of signal is 500 m, Television network, high speed, internet connection
Fibre optics Cable	Single mode, multi mode	100 gbps	Advantage) High speed) High bandwidth) High security Disadvantage.) Expensive) Requires Skilled) ~~Requires~~ Installation.	maximum distance of fibre optics Cable is around 100 meters

Result:

Thus, The different types of network Cable were studied successfully.

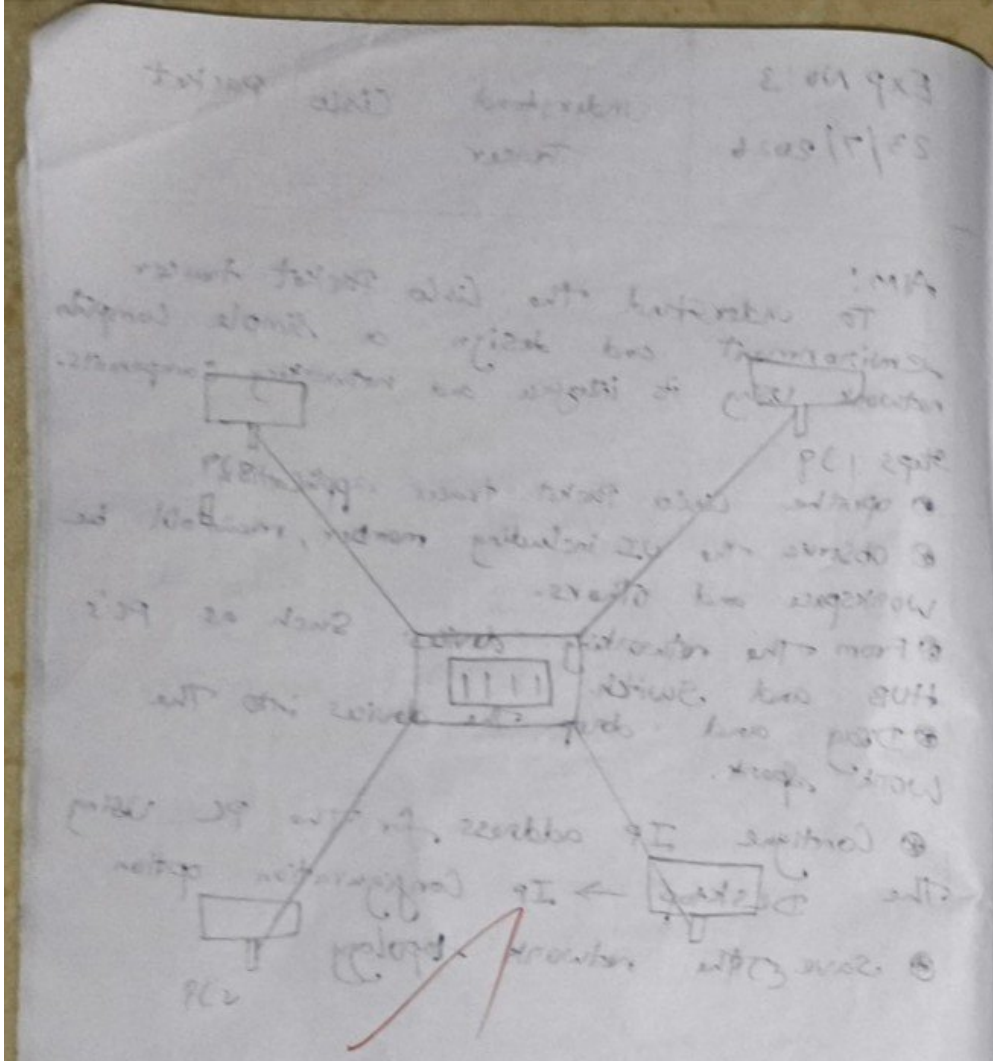
Exp No: 3
23/7/2026

Understand Cisco Packet Tracer

AIM:
To understand the Cisco Packet tracer environment and design a simple computer network using its interface and networking components.

Steps:
- open the Cisco Packet tracer application.
- Observe the UI including member, maintool be workspace and others.
- From the networking devices such as PC's HUB and Switch.
- Drag and drop the devices into the work space.
- Configure IP address for the PC using the Desktop → IP Configuration option
- save the network topology

Result:
The Cisco packet tracer environment was successfully and a simple IP Configuration and packet transmission was Sucessfully understood using both realtime and simulation mode.



Exp : 3.1
23/7/26

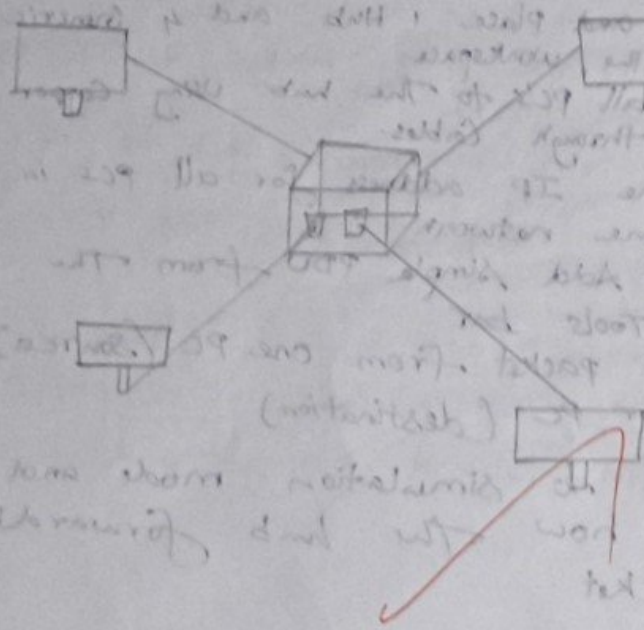
HUB

AIM:
To study the behavior of a Hub by observing how its forwards data packets using cisco packet tracer.

Steps:

- Open Cisco Packet Tracer
- During and Place 1 Hub and 4 Generic PC's into the workspace
- Connect all PC's to the hub using Copper Straight-through Cables.
- Configure IP address for all PCs in the same network
- Select Add simple PDU from the Common Tools bar
- send a packet from one PC (source) to another PC (destination)
- Switch to simulation mode and observe how the hub forwards the packet

Result:
The hub broadcasts the received packet to all connected devices. Every PC receives the packet but only destination PC accepts it, while the others discard it.



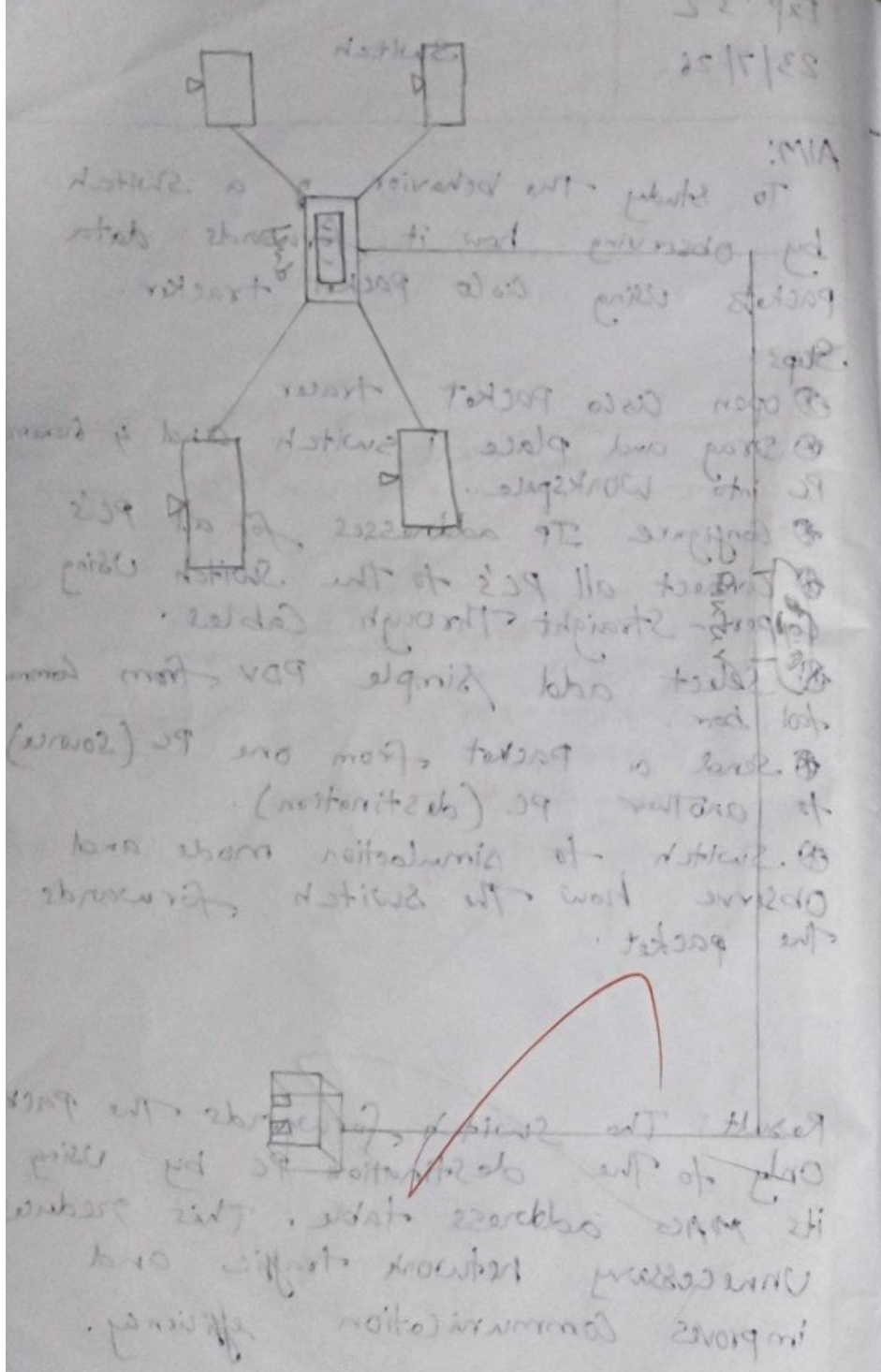
Exp : 3.2
23/7/26

Switch

AIM:
To study the behavior of a switch by observing how it forwards data packets using Cisco packet tracker.

Steps:
- open Cisco Packet tracer
- Drag and place 1 switch and 4 Generic PC into Workspace.
- Configure IP addresses for all PC's
- Connect all PC's to the Switch using Copper - Straight Through Cables.
- Select add simple PDV from common tool bar.
- Send a packet from one PC (source) to another PC (destination).
- Switch to simulation mode and observe how the switch forwards the packet.

Result: The switch forwards the Packet only to the destination PC by using its MAC address table. This reduces unnecessary network traffic and improves communication efficiency.



Exp: 3.3
23/7/26

HUB and Switch (Comparison)

Aim:

To Compare the packet forwarding behavior of a hub and a Switch Using Cisco packet Tracer.

Steps:

→ Create one network Using a hub and another Using a Switch, each Connected to 4 PCs

→ Configure IP address for all PCs in both network.

→ Send a simple PDU between two PCs in each network

→ observe packet transmission in Simulation mode

→ Compare how the hub and Switch forward packet.

Result:

The hub broadcasts packet to all connected devices, whereas the Switch forwards packet only to the intended destination.

Exp: 4
29/07/2026

Setup and Configure a LAN

AIM:

To Step and Configure a LAN Using Switch and ethernet Cables in our lab.

Procedure:

1) Plan and design a network topology taking into account network requirement and equipment location.

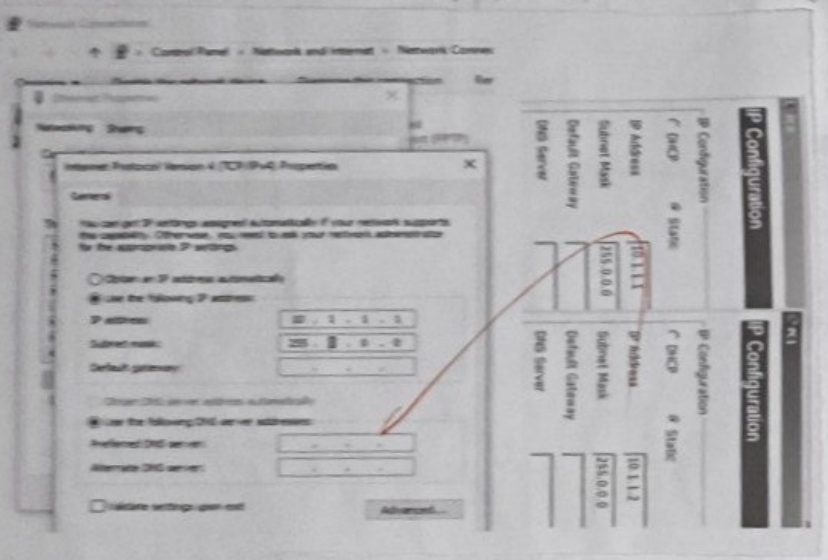
2) You can take 4 computer, a Switch & u or on port which in sufficient for network & These 20 port which is and 4 ethernet Cable.

3) Assign IP address to you pc's:

(i) Log on to the dint Computer admin & or a own.

(ii) Click network and Internet Connections.

(iii) Right Click (Top IPV4) → Click on Properties Select Use the following IP and assign IP address.



5) Configure a network Switch:-

i) Connect you Computer to the Switch using an Ethernet Cable.

ii) Configure basic setting - once your logged in you will be able to configure basic & Configure basic setting for the Switch.

Step 6: Check the connectivety between Switch and other machine by using P command in the Command prompt of the device.

Step 7: Select a folder → go to prop Click sharing tab → share it with everyone the same LAN

Step 8: They to access the shared folder from other Computer of the network.

80

Result:

The Program has been successfully executed.

Exp: 5 | Experiment on packet Capture: Wire Shark

20/08/2026

AIM:

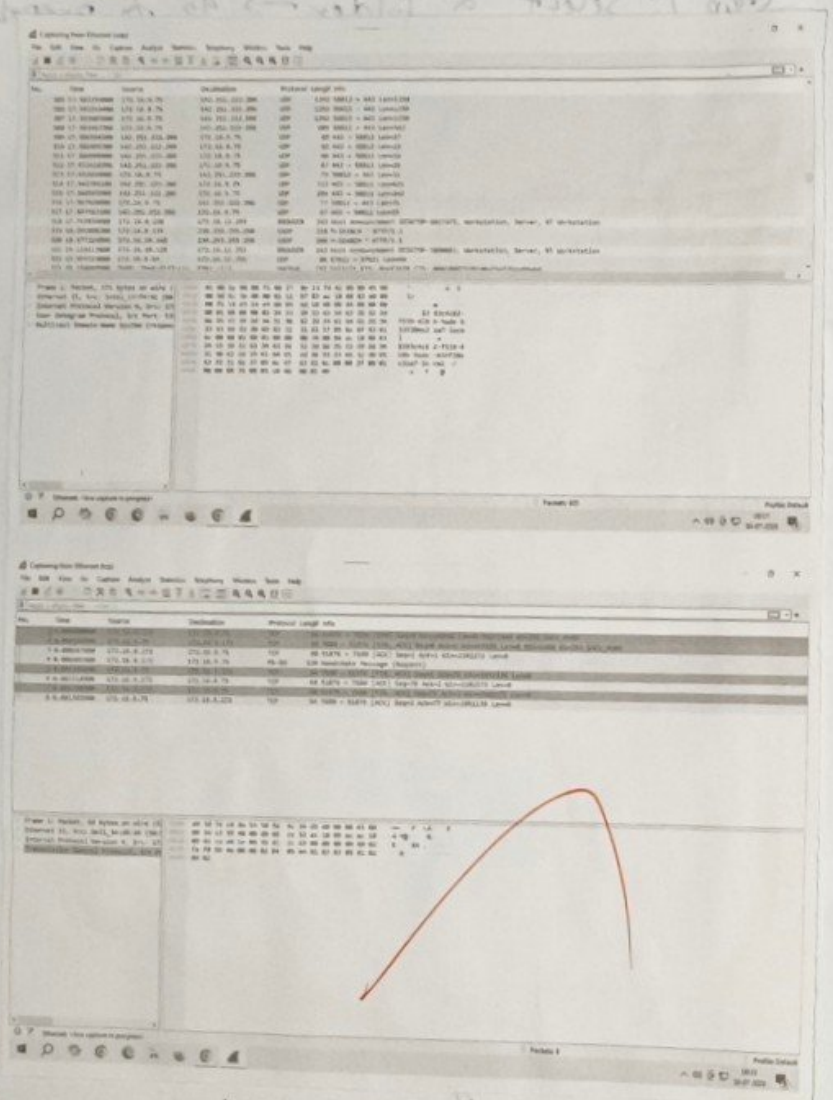
To perform a experiment on packet Capture Using wire Sharp tool.

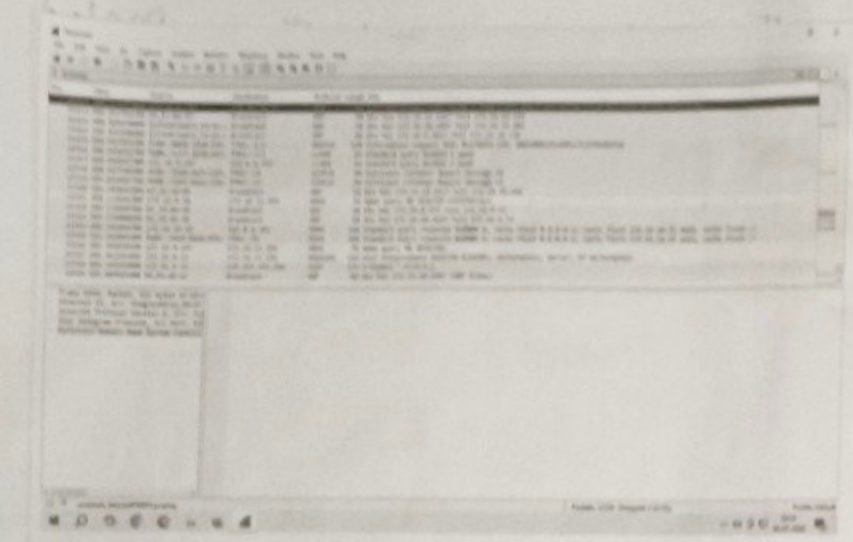
Procedure:

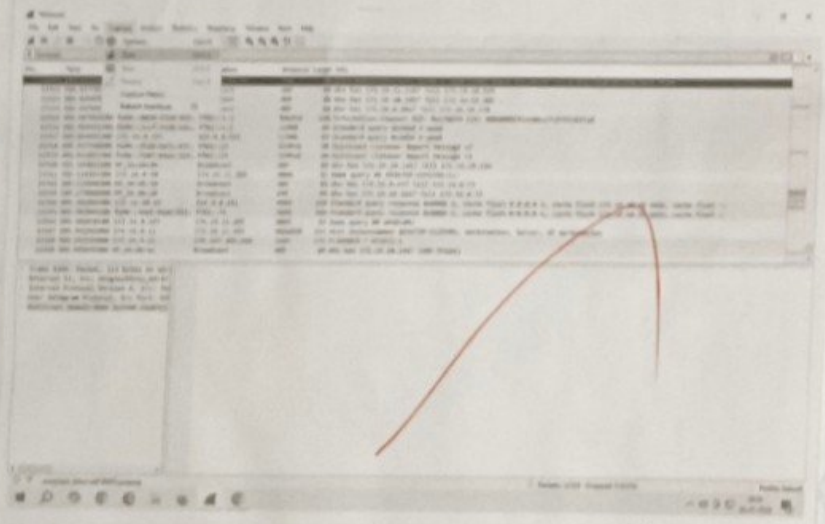
1) Create a filter to display only TCP/ODP packet, inspect the packet and provide the flow graph.
   - Select LAN in wireshark
   - Go to Capture option
   - Then click Start Capture
   - Then search TCP packet in Search bar.
   - Start the packet

2) Create a filth to display only ARP packet.
   - Go to Capture and Select Stop Capture.
   - The click Start Capture and search nep
   - Save the Packet.

3) Create and display is - filter only DNS packet.
   - Go to Capture and Select Stop Captu
   - Then click Start Capture and start DNS
   - Save the packet.







b) Create a filter to display only HTTP / ICMP DHCP and input the packet

- Select 'LAN' in wireShark
- Go to the capture options
- Select Stop Capture automatically after 100
- Search HTTP on IPI ICMP on DHCP on DHCP in the search bar
- Save the packet.

Result:

Thus, the Experiment on Capturing packet Using wiresheet tool tool has been unsuccessfully implemented.

block 1
Enter 4 data bits (space separated
eg: 1011): 1011

Original data bits: 1011
Encode Hamming Code (7 bits): 0110011
Enter bits position to introduce error
(1-7, 0: no error)
Transmitted Code with error added at
pos 3): 0100011

Received Code: 0100011
Corrected Code: 0110011
Recorded data bits: 1011

Exp No: 06
6/8/2026

Error Detection and Correction Using Hamming Code.

AIM:
To write a Program to implement error detection and correction using the Hamming Code.

Algorithm:
1. Start the Program
2. Read the number of 4 bit data blocks (n) forming the input data stream
3. a) For each 4-bit data block $D = d_1 d_2 d_3 d_4$ and i=4
b) Compute $P_1 = d_1 \text{ XOR } d_2 \text{ XOR } d_4$
$P_2 = d_1 \text{ XOR } d_3 \text{ XOR } d_4$
$P_3 = d_2 \text{ XOR } d_3 \text{ XOR } d_4$
4) Compute error position = $C_4 C_2 C_1$
5) if Error Position = 0, no error has Correct.
6) Extract the bit at "Error position" is inverted to correct the error
7) Stop the Program.

Program:

```
def chCode(d)
    """d: list of 4 bits [d1 d2 d3 d4]
    - return 7-bit hamming code list
h = [0]*7    h[0] = h[2]^h[4]^h[6]
h = [2] = d[0]    h[1] = h[2]^[5]^h[
```

- - Block 2 - -

Enter 4 data bits (space separated) eg: 1011)

Original data bits: 0110

Encoded Hamming Code (7 bits: 0110011

Recieved Code: 1100100

Checking for error...

Corrected Code: 1100110

Restored data bits: 0110

```
h[4] = d[1]        h[3] = h[4] ^ d[3] ^ h[5]
h[5] = d[2]
h[0] = d[3]
if error_pos != 0
  Print(F"Enter detected at bit
position of error: pos -> Correcting...")
else:
  Print("No error detected")
  return error_pos
For b in range(1, n+1)
  Print(F"\n--- block {b} ---")
  d = list(map int, input("Enter 4 data
bits)
  n = encode(d)
  Print(f"encoded data bits)
  Print(f"encoded Hamming Code {to_str}")
  Print(f"extended Code: {to_str(n)}")
  Print(F"Corrected Code: {to_str(n4)}
```

Result:

The Program to implement error detection and correction using the Hamming Code Corrupt was successfully written in Python.

Enter total number of frame to send : 4
Enter Sending Window Size : 4
Enter frame number to Simulate a lost
Lineed Window : [Frame 0 - Frame - 3]
Node A → Sending Frame 0
Node A → Sending Frame 1
Node A → Sending Frame 2
Node B → Recieved Frame 0, Sending table
Node B → Recieved Frame, Sending Ack,
Node B → Recieved Frame 2, Seding Ack 2
SenAer window : [Frame 4 - Frame 7]
Node A → Sending Frame 4
Node A → Sending Frame 5
Node A → Sending Frame 6
Node A → Sending Frame 7
Sender Window [Frame 4 - Frame 7]
Node A → Sending Frame 5
Node A → Sending Frame 6
Node A → Sending Frame 7
Node B → Received Frame 4, Sending Ack 4
Node B → Received Frame 5 Sending Ack 5
Node B → Received Frame 7, Sending Ack 7

Exp No: 07 | Flow control at data unit layer
6/8/2026 | Using sliding window Protect

AIM : To write a program to implement flow control at the data wik layer using the sliding window protocol, and to simulate the flow flame to be send (N) senders window from one node to another

Algorithm :
1. Start the Program
2. Read the total Number of frame to be Send (N), Senders window Size (W).
3. Initialize base = 0 and next Sq = 0
4. Repeat while base < N
a. Determine the current sender wind
b. it the Lost feame lies within the feame.
C. The recruiter dis Cards the Lost frame and Subsequent.
d. Else no loss in this batch.
(i) The recruita accepts each frame in order and sends Cumlative.
5. Repeat step 4 until all N frame have been transmitted acknowledge
6. Stop the Program.

Program :
def Simulated (total - frame, window.x
base = 0

```
While base < total_frames
    Window < end = min (base + window_size - 1,
                        total
                        frames - 1)
if lost_frame is none and base.< = lost
  frame < next_seq :
  Print (f" node B -> Frame {lost_frame}
  Print (f" Node A -> Timeout {lost_frame}
     next_seq = lost_Frame
     lost_frame = xlone
for f in Range (base, next_seq):
   Print (f "Node B -> Recial Frame {F},
     Sending Ack (f)"
     base = next_seq
def main ();
  total_frame = int (input (F"Enter
  frame, number to simulate as lost /n:")
  if lost_Frame = -1:
     lost_Frame = xlone
  Simulate (total_Frame, window_size,
     lost_frame)
  if. name - = - main - ";
  main {}
```

(8) Result:

The Program to implement flow control The data link layer using the Slidey Window Protocal was successfully written in Python and verified by simulating the flow of frame b/w two nodes.

Ex.No:09 SUBNETTING IN CISCO PACKET TRACER

AIM:
Implementation of SUBNETTING in Cisco Packet tracer similator.

·) classless IP subnetting is a technique That allows for more efficient use of IP addresses by allowing for subnet masks that are not just the default masks for each IP class.

* Creating a network topology.
→ The first step - to create a network topology is packet Tracer
→ The blank network topology that we can use to add devices

1) Adding the devices
→ Router witches and PCs
→ Connect the devices by dragging a cable from one devices port to another.

* Subnetting:
→ Network address of 192.168.1.0/24 to provide enough space for at least 5 addressies.
→ Use a/27 Subnet mask. This will give us 8 subnets with 30 host addresses each.

* Plan the IP addressing:
→ Start with network 192.168.10/24.
→ A/27 Subnet provides
→ 32 total address
→ 8 Subnet

→ Configure Router R1
Enable
Configure terminal

→ Configure the first interface.
interface Gigabit Ethernet 0/0

no shutdown
exit
→ Configure the second interface
interface gigabit Ethernet 0/0
ip address 192.168.2.1 255.255.255.224
no shoutdown
exit
→ Exit with gration mode:
end
→ Configure Router R2 mode:
enable
Configure terminal
→ Configure the first interface.
interface fastEthernet 0/0
ip address. 192.168.3.1 255.255.255.224
no shutdown
exit
→ Configure the second interface.
interface Fast Ethernet 0/1
ip address 192.168.4.1 255.255.255.224
no shutdown
Exit.
→ PCs Connected to S1

PC1	192.168.1.11	255.255.255.224	192.168.1.1
PC2	192.168.1.12	255.255.255.224	192.168.1.1
PC3	192.168.1.13	255.255.255.224	192.168.1.1
PC4	192.168.1.14	255.255.255.224	192.168.1.1
PC5	192.168.1.15	255.255.255.224	192.168.1.1

→ R2 Fo/0
→ R2 Fo/1

Router Configuration.

enable
configure terminal
interfaces Fast Ethernet 0/0
ip address {IP address} {subnet mask}
no shutdown
exit
interface Fast Ethernet 0/1
ip address {IP address} {subnet mask}
no Submission
exit
→ Configure the Router.
Right click the Router - CLI
Enter following Commands
enable
configure terminal
interface Fast Ethernet 0/0
ip address {IP address} {subnet mask
no shutdown
Exit
→ Configure the Switch:
* Right Click Switch → CLI
enable
Configure terminal
interface Fast Ethernet 0/1
Switch port mode access
Exit.